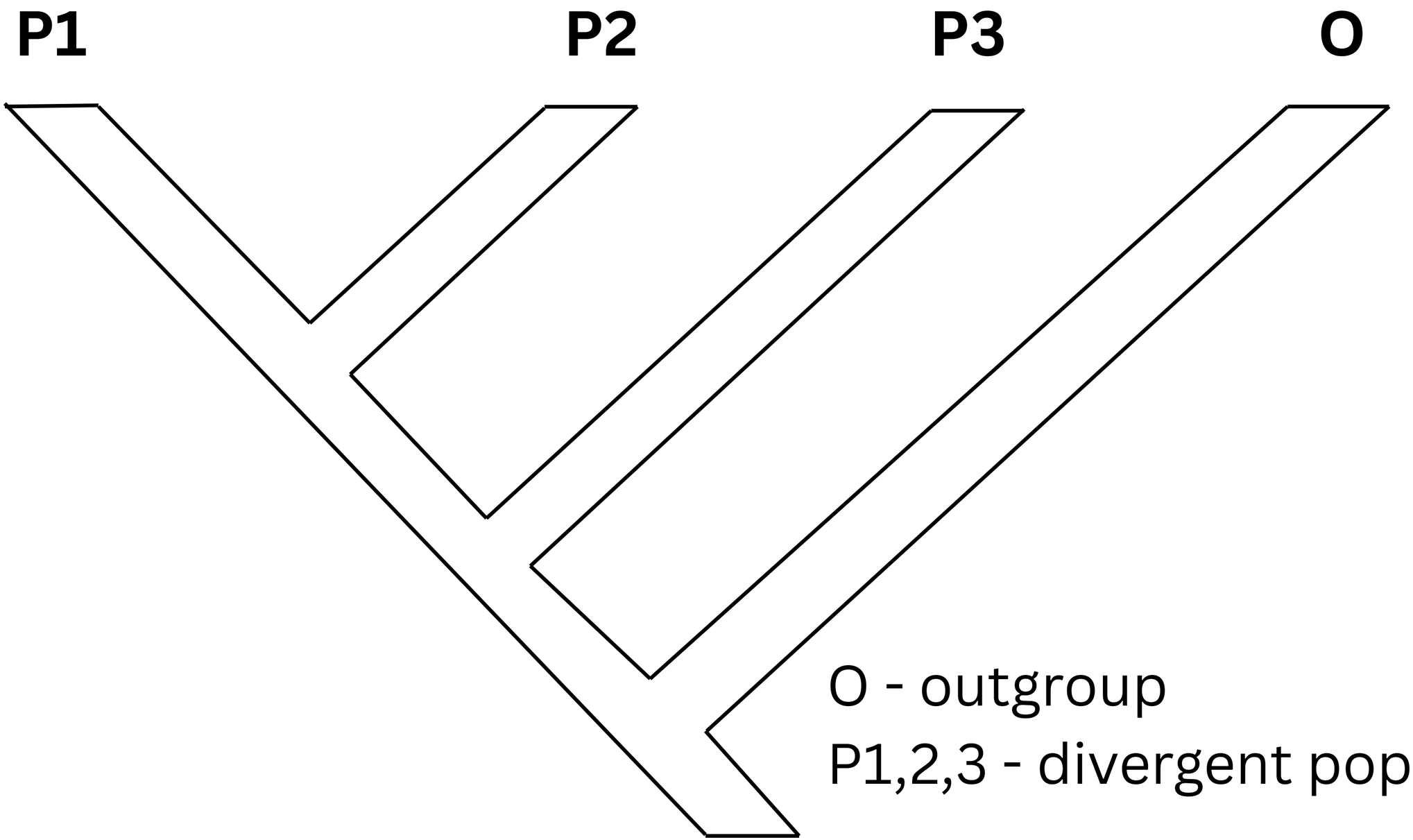
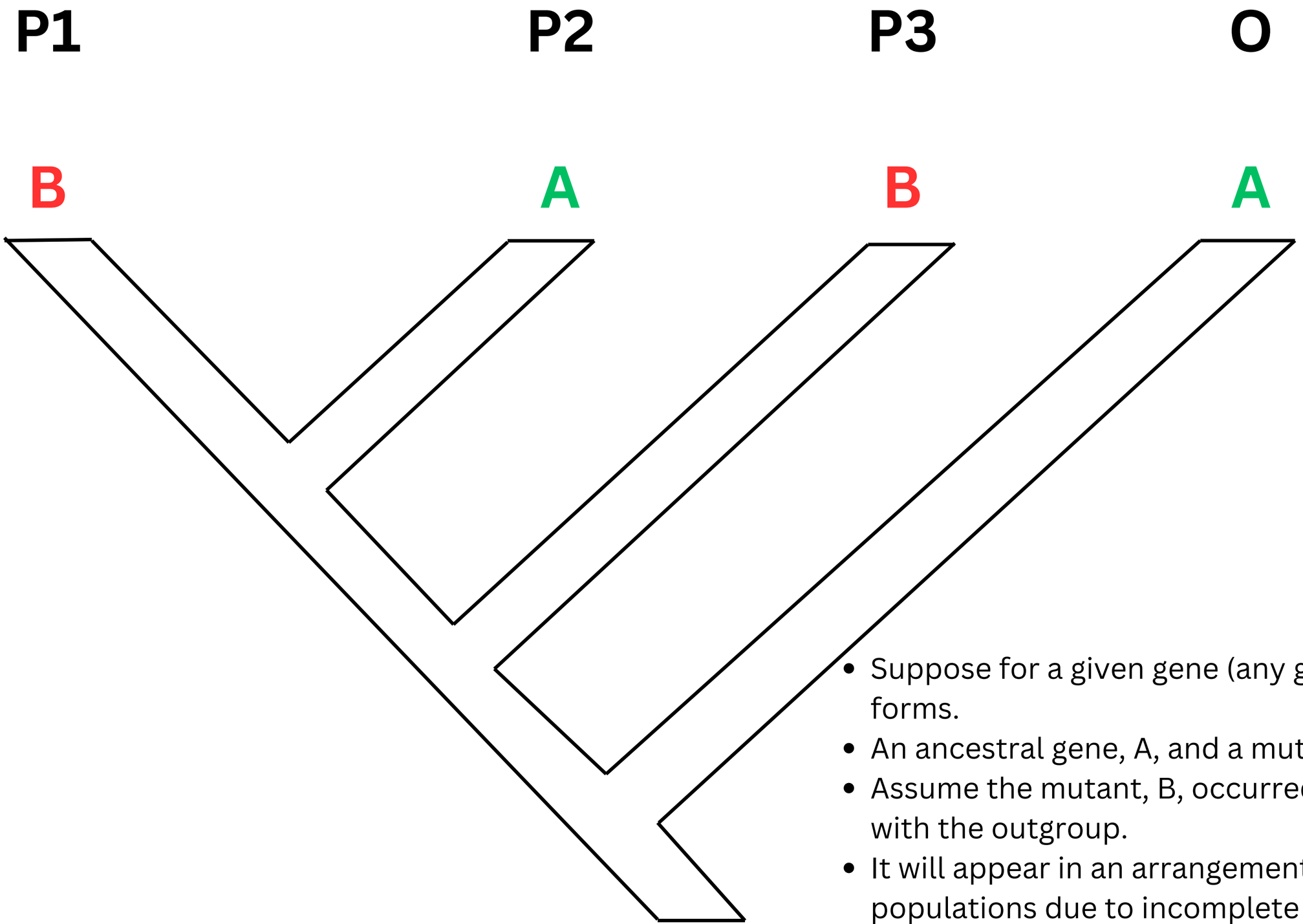
ABBA-BABA (The D-statistic)

Imagine a phylogeny of populations

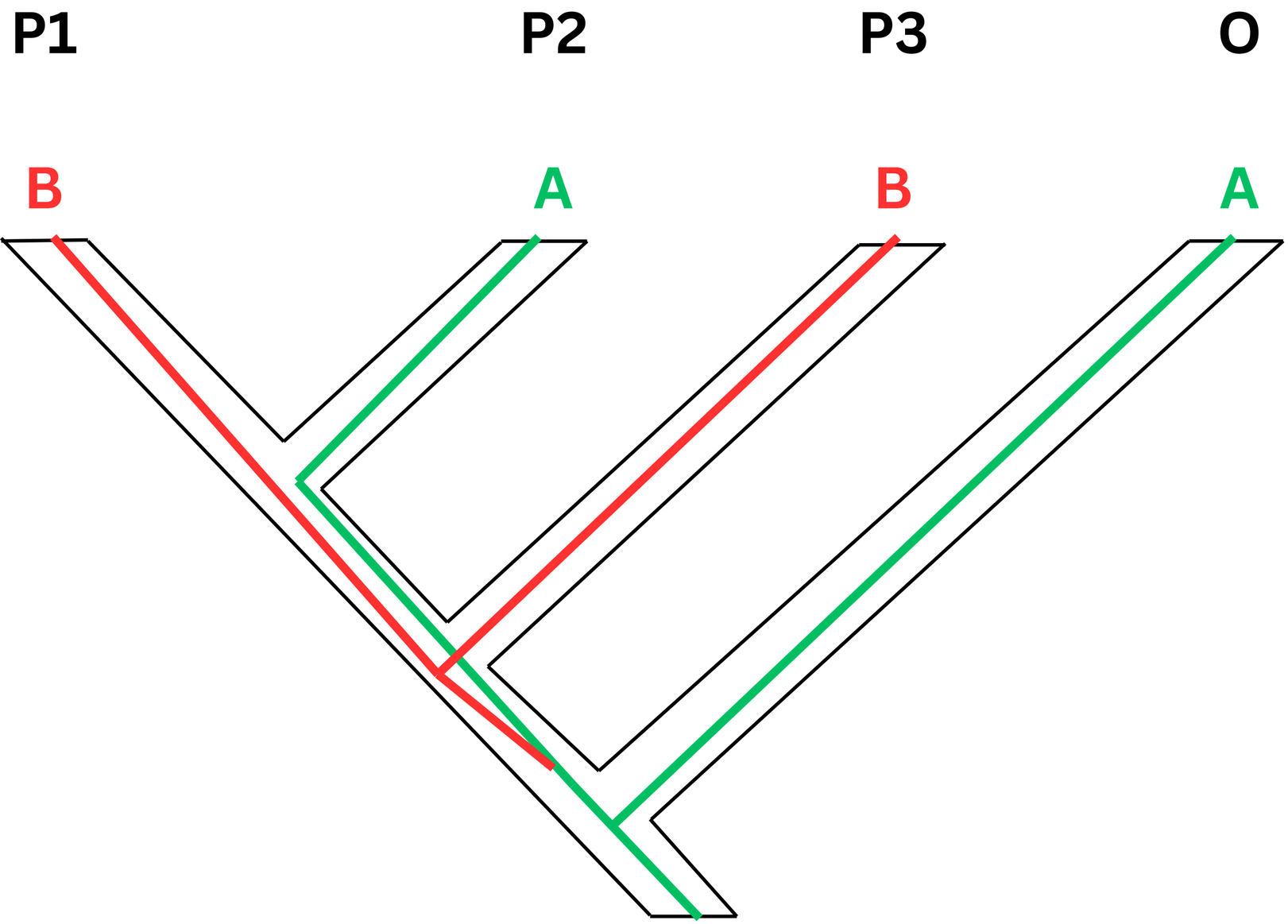


O - outgroup

P1,2,3 - divergent populations



- Suppose for a given gene (any gene) there are two forms.
- An ancestral gene, A, and a mutated gene, B.
- Assume the mutant, B, occurred after the split with the outgroup.
- It will appear in an arrangement in the subsequent populations due to incomplete lineage sorting.

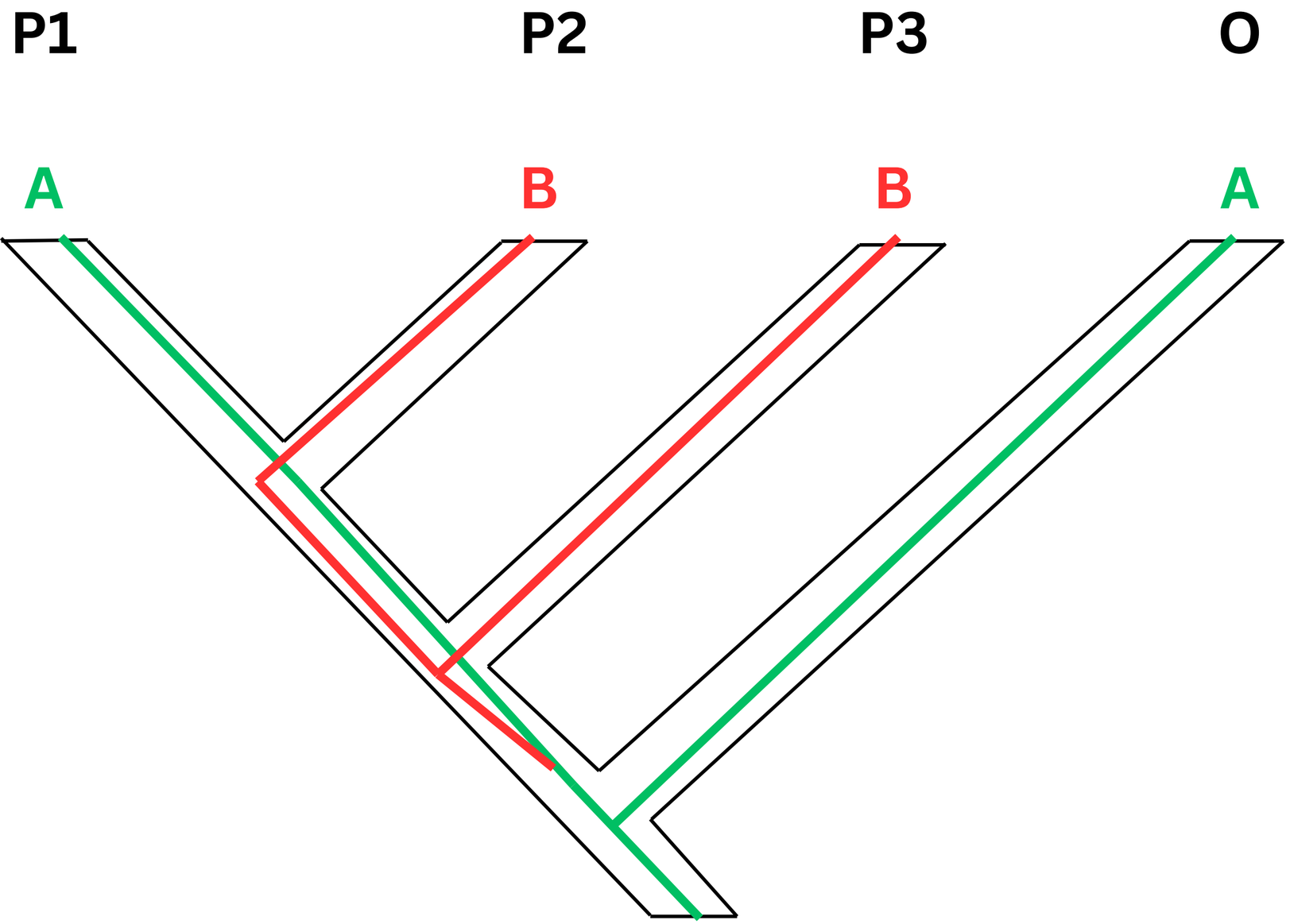


Incomplete Lineage Sorting (ILS):

- Process where two genes diverge within an ancestral population before that population splits and diverges itself.
- Results in gene tree not matching the species/population tree.

In this example:

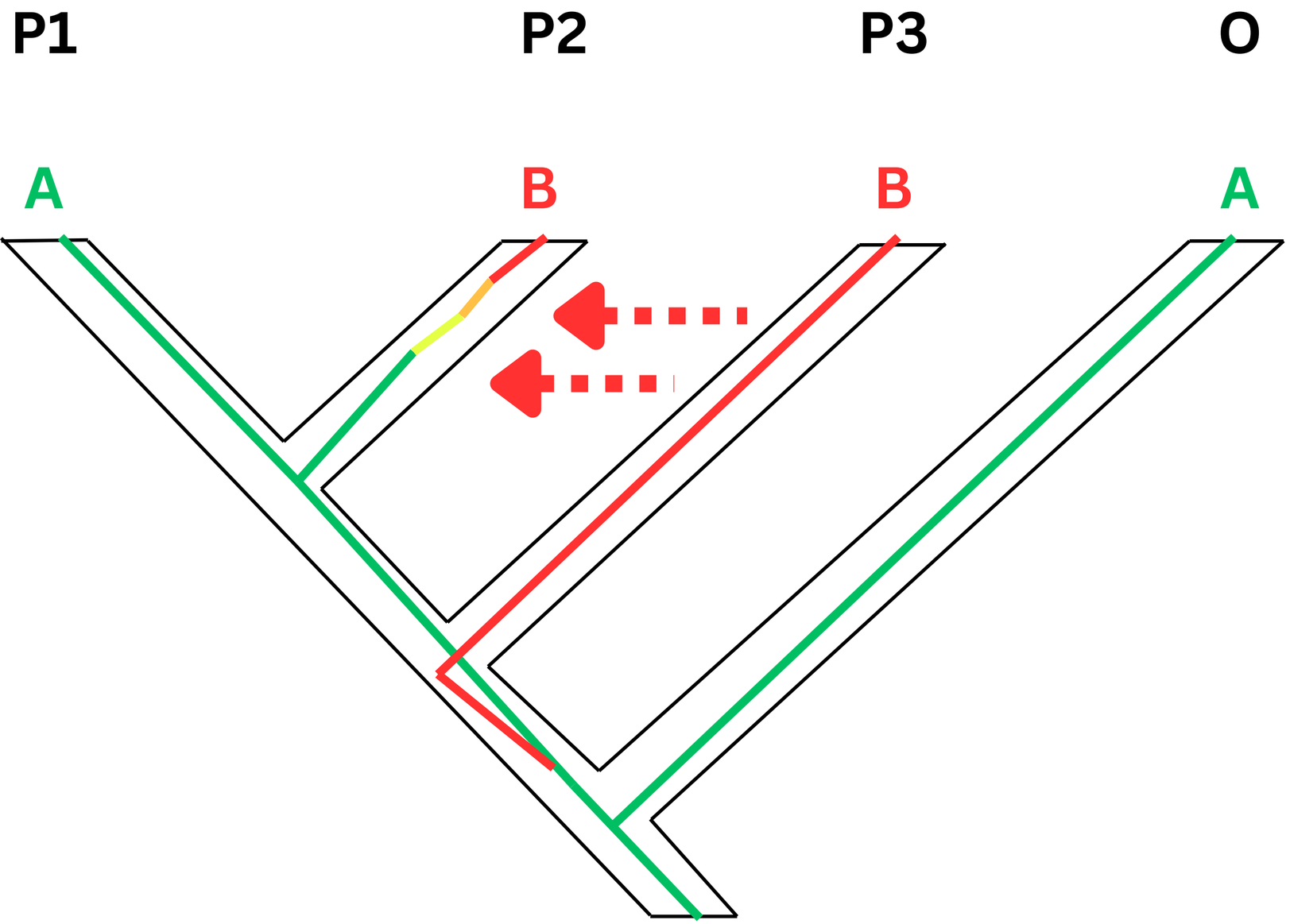
- Genes A and B diverge after the split with the outgroup.
- Both genes remain within the ancestral population as alleles.
- Through drift population P3 inherits the mutant allele.
- Both alleles remain in ancestral population of P1 and 2.
- P2 inherits the ancestral allele while P1 inherits the mutant allele via drift.



In this example:

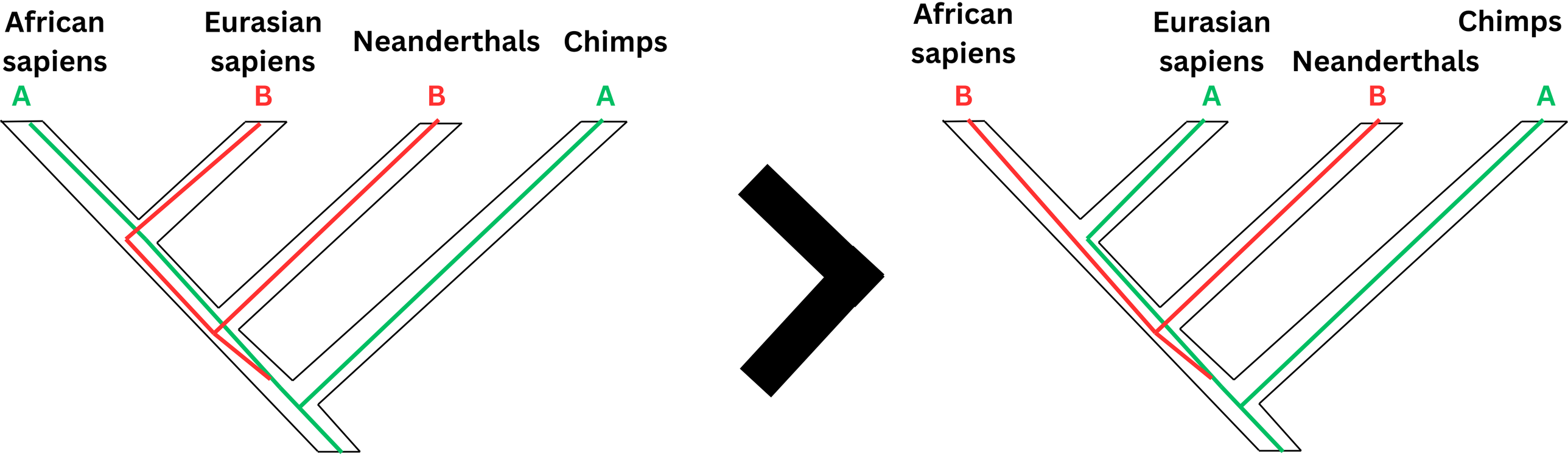
- Equally the arrangement of the two alleles could be switched to a ABBA system instead.
- Nothing has fundamentally changed.
- Drift (incomplete lineage sorting) just caused P1 to inherit gene A instead of B and vice versa for P2.

- If you were to pick 100 genes at random that had mutant strains (B strain alongside A) you would expect 50 to follow the ABBA pattern and 50 to follow the BABA pattern.
- This is because of incomplete lineage sorting/random forces shaping the distribution.
- I.E., the ratio of ABBAs to BABAs is 1.



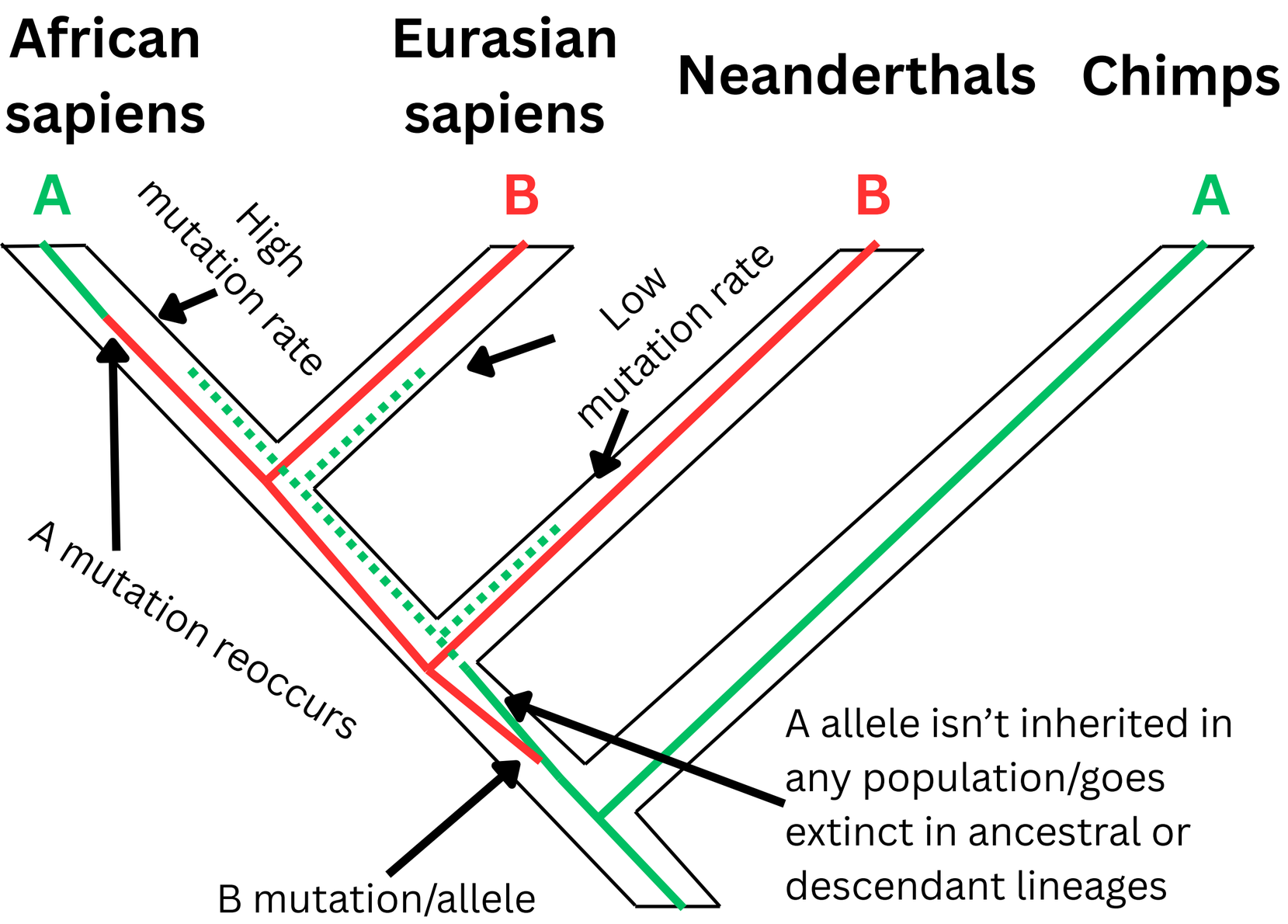
Introgression:

- In this hypothetical system there is no incomplete lineage sorting giving either ABBA or BABA.
- Instead the genes are coming from introgression.
- If introgression did occur between any populations in this tree it would upset the 50/50 balance of alleles you get from incomplete lineage sorting.
- Instead, in this case, it'd increase the number of ABBAs.
- This deviates the ratio of ABBAs to BABAs away from 1 - this is what indicates introgression.
- ANY deviation is considered a signal of introgression.
- In this specific example it's ABBA>BABA.



Context in human evolution:

- If you replace P1,2,3 and O with the following human groups you can perform ABBA-BABA ratio (D-statistic).
- You see a higher ratio of ABBAs to BABAs doing this, indicating a deviation from incomplete lineage sorting and suggesting introgression between neanderthals and eurasian sapiens.



PROBLEMS (Bill Amos)

- The D-statistic **ASSUMES** a **constant** mutation rate along all lineages **AND** that **ALL** arrangements are due to INCOMPLETE LINEAGE SORTING.
- This is often not the case - as shown in the figure.
- Africans have a higher mutation rate than out of africa populations due to being more heterozygous - Bill Amos.
- This means the African lineages have a higher probability of mutating away from an ancestral B mutant, back to the ancestral gene seen in the outgroup.

- Thus, giving the illusion of introgression.
- Mutations back to an ancestral state are also more likely as A->T or G->C and vice versa are more likely than pyramidine ->purine mutations.